

=== Generated Equipment Data ===

Temperature: 78

Pressure: 78

Vibration: 8.0

Voltage: 200

=== Validation Results ===

Temperature: VALID

Pressure: VALID

Vibration: VALID

Voltage: VALID

=== Diagnostic Results ===

Total: 364.0

Average: 91.0

Highest: 200

Lowest: 8.0

Status: WARNING

=== Execution Log ===

generate_readings: SUCCESS

validate: SUCCESS

calculate: SUCCESS

classify: SUCCESS

=== Generated Fault Data ===

Surname: SIRUNO

SEED_NUM: 8

Favorite Artist: BINI

Fault Code: 4610

=== Recursive Trace ===

Call 1: 4610 -> 11

Call 2: 11 -> 2

Call 3: 2 -> ROOT FAULT

=== Number of Recursive Calls ===

3

=== Execution Log ===

generate_readings: SUCCESS

validate: SUCCESS

calculate: SUCCESS

classify: SUCCESS

generate_fault_code: SUCCESS

trace_fault: Call 1

trace_fault: Call 2

trace_fault: ROOT FAULT

=== Final Output ===

Fault Code: 4610

Root Fault: 2

Recursive Calls: 3

```
=== Student Specific Inputs ===  
Surname: SIRUNO  
SEED_NUM: 8  
Favorite Artist: BINI  
  
=== Generated Telemetry Data ===  
81.9  
81.9  
8.4  
210.0  
60.9  
31.5  
105.0  
  
=== Valid / Invalid Results ===  
Valid Readings: 7  
Invalid Readings: 0  
  
=== Processed Results ===  
Processed: 81.9  
Processed: 81.9  
Processed: 8.4  
Processed: 210.0  
Processed: 60.9  
Processed: 31.5  
Processed: 105.0  
  
=== Detected Abnormal Conditions ===  
LOW VALUE: 8.4  
HIGH VALUE: 210.0  
  
=== Recursive Analysis ===  
Analyzing: LOW VALUE: 8.4  
Analyzing: HIGH VALUE: 210.0  
Analysis complete
```

```
=== Final Diagnostic Summary ===  
Processed Readings: 7  
Valid Readings: 7  
Invalid Readings: 0  
Abnormal Conditions: 2  
Overall Equipment Status: WARNING  
  
=== Execution Log ===  
detect_abnormal: STARTED  
detect_abnormal: SUCCESS  
  
=== Final Output ===  
Equipment Status: WARNING
```